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An LLM and Knowledge-Graph Based Escalation Engine for Root-Cause Analysis in Advanced Robotics Systems

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***Abstract*—Advanced robotic systems involve the integration of tightly coupled mechanical, sensing, control, firmware, networking, and site-level components. As a result, diagnosing and escalating problems is difficult. When failures occur, support teams struggle to convert scattered field data into engineering-quality escalations. This problem occurs frequently when there are delays in escalating issues, poor communication among different types of troubleshooting expertise, and inconsistencies in identifying root causes. To resolve this issue, this paper presents a knowledge-grounded escalation engine. The engine leverages historical maintenance information, a robotics-specific knowledge graph, and a large language model (LLM). The system will structure disparate field data, locate relevant graph evidence and past incidents similar in nature to the current issue, and use the LLM to create ranked lists of possible root causes for the problem. It also drafts a structured escalation package. Ultimately, this proposed methodology will increase the diagnostic quality of the robot-generated solution, reduce the time to escalate an issue, produce more uniform escalation package content, and document clear evidence trails for each escalation.**

***Index Terms*—Advanced robotics maintenance, root-cause analysis, escalation systems, knowledge graphs, retrieval-augmented generation, GraphRAG, human-in-the-loop decision support, industrial AI risk management.**

I. Introduction

Root-cause analysis in advanced robotics is difficult because faults often emerge from interactions among mechanical, sensing, control, and software layers, and these interactions can vary across sites due to differences in environment, procedures, maintenance skill, and software versions [1]. At the same time, industrial escalation is hindered by unstructured field data such as technician notes, inconsistent abbreviations, alarm codes, and loosely formatted logs, while engineering teams require concise, evidence-based escalation packets with failure context, reproduction steps, and likely causes; this mismatch often leads to delays and inconsistent diagnoses that depend heavily on individual judgment and experience [2].

To address these ongoing challenges, this paper proposes an escalation engine that combines structured failure knowledge with LLM-based generation. Two findings justify this direction. First, knowledge graphs digitally capture maintenance knowledge, enabling querying and reasoning to support decisions and fault diagnosis [2]. Second, while large language models can generate summaries and reports, they may produce inaccuracies or misleading statements, which recent research terms "hallucination." This underlines the importance of grounded retrieval, uncertainty detection, and constraint generation [3].

A. Problem Statement

Current escalation workflows rely heavily on individual expertise and make limited use of historical case knowledge, much of which remains buried in unstructured records such as free-text maintenance notes [4]. This work addresses that gap by developing an intelligent escalation assistance tool that structures historical maintenance knowledge, incorporates fault-specific context, identifies likely causes, and preserves a traceable decision record. The objectives are to build a robotics fault knowledge graph, integrate it with LLM-based generation to produce evidence-grounded hypotheses and draft escalations, and evaluate whether this approach improves diagnostic quality, completion time, and user satisfaction relative to manual methods and non-grounded baselines [2].

B. Research Gap

This paper addresses the gap related to the lack of integrated, explainable escalation support for robotic maintenance operations that (i) provide a ranking of possible reasons for probable faults, (ii) produce escalation-ready communication products, and (iii) establish each major assertion through structural relationships and past historical evidence. Although prior studies on robotics fault detection and diagnosis provide effective mechanisms for detecting and identifying faults, they do not produce an escalation draft [5]. Prior studies on maintenance knowledge graphs have organized the knowledge required for troubleshooting; however, they typically provide static information or are not incorporated into the operationalization process in an escalation workflow [1], [4].

C. Contributions

The main contributions of this research are fourfold. First, it develops a domain-specific knowledge graph schema for

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robotics maintenance reasoning and escalation tasks, and second, it introduces an escalation workflow that integrates graph-based retrieval of historical cases with retrieval-augmented LLM generation. In addition, the work presents a retrieval-augmented reasoning pipeline that generates ranked hypotheses with evidence trails, along with a formal evaluation protocol and instrumentation plan to measure both diagnostic and operational impacts of the proposed knowledge-graph-based approach [6].

II. Related Work

This research builds on prior work in robotics fault diagnosis, industrial maintenance knowledge graphs, and retrieval-augmented large language models, while addressing a gap in integrated escalation support for robotic maintenance operations. Existing robotics fault detection and diagnosis methods include structured approaches such as FMEA and FMECA, as well as model-based and data-driven techniques; however, prior studies have shown that robotic systems remain especially difficult to diagnose because of their autonomy, sensor dependence, dynamic operating context, and multi-degree-of-freedom behavior [3], [4], [6]. As a result, these approaches are valuable for detecting and explaining faults, but they generally do not produce structured escalation drafts for engineering teams.

A second relevant area is the use of knowledge graphs(KGs) in industrial maintenance. Prior work has shown that KGs can support maintenance decision-making by organizing entities, faults, actions, and outcomes into structures that enable querying and reasoning [5], [6]. Research has also shown that although maintenance data is often unstructured and difficult to reuse, graph-based representations combined with ontology-based reasoning can improve retrieval and pattern discovery, and can link disparate fault-related information into more usable diagnostic knowledge [6], [7]. However, most existing maintenance KGs remain either expert-heavy, limited in scope, or disconnected from the operational escalation workflows where engineers need evidence-backed fault summaries and follow-up recommendations.

A third area is the use of LLMs and retrieval-augmented generation(RAG) for technical decision support. While LLMs are effective at summarization and report generation, their outputs may contain hallucinations, making grounded retrieval essential in high-stakes engineering environments [7]. RAG improves factuality and inspectability by grounding generation in retrieved external evidence, and recent graph-based extensions such as GraphRAG have shown that relational structure can improve retrieval precision and support multi-step reasoning [5], [7]. Nevertheless, there is still limited work that combines LLM-based drafting with a domain-maintained robotics fault knowledge graph and an operational pipeline capable of producing traceable, escalation-ready engineering outputs rather than generic explanations.

III. System and Methodology

This part includes the architecture model, data flow, knowledge representation, retrieval/prompts, and the output format for the escalation process. Reproducibility and operational transparency are important—what inputs come in, what is retrieved, what is generated, and what constraints are placed upon outputs to limit the potential for unsupported reasoning.

A. System Overview

The Escalation Engine will take data regarding one or multiple robot assets/fleets, normalize that data into many different forms of entity/event records, update a knowledge graph associated with robotics, retrieve information from the knowledge graph, create a contextually relevant information package for an LLM, and the LLM will produce: label=i)

1) ordered lists of possible cause hypotheses;
2) an evidence chain linking symptoms/alarms/components with historical failure;
3) a draft escalation package that complies with the engineering requirements.

This format uses the established RAG principle, combining parametric generation with external evidence to increase the likelihood of factuality and inspectability.

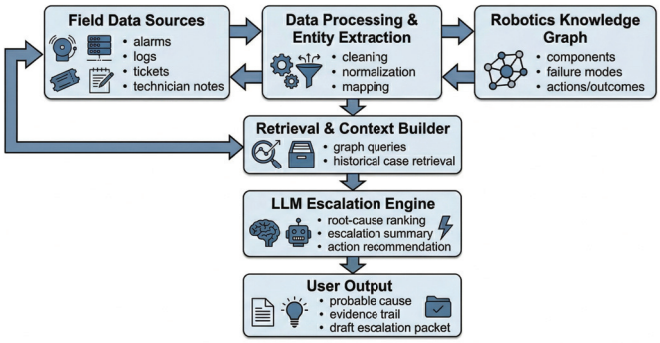


Fig. 1. Overall System Architecture.

B. System Components, Data Sources, and Knowledge Graph Design

The system operates in five stages. First, field data sources provide heterogeneous evidence and reflect the fragmentation common in robotics maintenance, where much of the data remains unstructured and difficult to analyze [8]. Second, data processing normalizes this information for consistent insertion and retrieval in downstream graph construction, following established KG pipeline practices for integrating multi-source data [4]. Third, the robotics knowledge graph captures both system structure and failure knowledge, enabling traceable querying and reasoning for maintenance support [6]. Fourth, retrieval and context creation use graph structure to assemble multi-hop evidence for complex fault scenarios, consistent with GraphRAG goals of improving retrieval precision and contextual relevance [9]. Finally, the LLM escalation engine

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generates ranked hypotheses and draft escalations grounded in retrieved evidence, reducing the risk of hallucinated outputs [5].

C. Data Sources and Preprocessing

The escalation engine uses common operational and maintenance data from robotics deployments. A canonical schema includes alarms and fault codes from controllers and safety systems, operational logs from firmware and control software, tickets and maintenance reports from CMMS or ticketing tools, histories of part replacements and repair outcomes, and quality observations related to recurring defects. Because these records are often unstructured, the pipeline applies text normalization and entity extraction to improve reuse and retrieval [7], [10].

Preprocessing includes removing duplicate tickets, cleaning subsystem names and fault codes, aligning timestamps within a reasonable tolerance, linking events to asset identifiers and operating configurations, and applying anonymization where needed. These steps are consistent with prior work on heterogeneous source integration, provenance, quality assurance, and KG construction pipelines [11].

D. Knowledge Graph Design

The knowledge graph acts as a reusable diagnostic memory that links symptoms, alarms, components, corrective actions, and outcomes across incidents and sites. By making these relationships explicit, it supports querying, reasoning, and traceable decision-making while reducing reliance on implicit expert knowledge [2]. The graph may be implemented as either a property graph or an RDF dataset. Property graphs support flexible edge properties and traversal-based queries, while RDF and SPARQL provide standardized semantics and expressive querying across sources [4]. The framework is representation-agnostic as long as retrieval returns a compact evidence subgraph with typed relationships and provenance.

1) Entity Definition: The entity design mirrors maintenance reasoning in robotics. It includes robot models and asset instances, major subsystems, individual components, fault and alarm codes, observed symptoms, maintenance actions, software and firmware versions, site conditions, and incident outcomes. This structure supports reuse of experience across past incidents by linking symptoms, subsystems, actions, and outcomes in a form that can be queried and reasoned over. Prior PHM research in aviation similarly shows that KGs can connect disparate fault data and support diagnosis and operator assistance [11], [12].

2) Relationship Modeling: The relationship design supports traceable paths from evidence to plausible causes. For example, a robot model may contain a component, an alarm may indicate a fault mode, and a symptom may be associated with a component or failure mode. Site conditions may contribute to symptoms, maintenance actions may resolve outcomes, and temporal relations may capture event order across alarms and symptoms. Additional relations can record where an incident occurs and which team it is escalated to. These links are important because graph structure enables multi-hop retrieval and richer contextual reasoning than isolated document retrieval alone [9].

TABLE I
KNOWLEDGE GRAPH RELATIONSHIPS FOR ROBOTICS MAINTENANCE

Entity Type	Example	Relationship	Target	Purpose
Robot Model	Drive Unit X	contains	Motor Controller	Represents system structure
Symptom	Irregular vibration	associated_with	Bearing wear	Links observations to faults
Alarm Code	A145	indicates	Encoder fault	Maps machine alarms to causes
Maintenance Action	Replace belt	resolved	Tracking drift	Stores corrective history
Site Condition	High dust	contributes_to	Sensor contamination	Captures environment effects

The schema integrates both structural information about the machine's physical layout and procedural knowledge related to equipment failures. Smart manufacturing surveys have emphasized the need for semantically-oriented models that can effectively model the complexity of a given production system and enable informed decision-making [13].

E. LLM-Assisted Escalation Engine

The LLM does not serve as an unstructured diagnostic tool. Rather, it serves as a structured generator/drafter tool after evidence has been retrieved through the structured query. This approach limits reliance on unreliable aspects of LLM generation, i.e., hallucinations, and aligns with evidence-driven generation and detection research [10].

1) Prompting and Retrieval Strategy: The pipeline creates a structured prompt that consists of four segments, which define the search space for evidence:

- A normalized version of the incident summary, including alarm messages and symptom descriptions.
- A compact, graph-based representation of a sub-graph within the entire system (as an example) using typed triple-sets or edge lists referencing the original log entries or tickets.
- The pipeline also uses similar historical cases retrieved from previous incidents and resolutions, along with additional structural fields such as "actions taken" and "status." Research on maintenance knowledge graphs indicates that graph-based, structured representations of data may offer benefits for both information retrieval and pattern identification. This may be particularly true if paired with ontology frameworks and/or reasoning tools [14].
- Finally, the pipeline defines an escalation template providing rules for the expected input fields. Best-practice escalation procedures typically require reliable, consistent handoffs of work. Therefore, establishing transparent documentation and accounting for the reliability and consistency of workflows is also recommended to manage AI risks [7].

Graph retrieval may be performed in two stages. In the first stage, candidate root causes are identified within a small number of graph neighborhoods surrounding observable symptoms and alarm codes via a depth-limited traversal. In the second stage, a re-scoring stage evaluates the relevance of

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these candidates using a weighted sum of textual similarity to the incident description and relational support based on the presence of paths connecting evidence in the graph. An elementary form of a deterministic scoring function may be written as follows:

$$s(c \mid x) = \alpha \,\mathrm{sim}(x, c) + \beta \,\mathrm{path}(E_x, c) + \gamma \,\log p_0(c)$$

where x describes the information available about an incident, c is one of the potential explanations for that incident, sim is a way to compare the words of the incident description with other words associated with the explanation(s), path provides a measure of the relationship between evidence E_x to candidate c, and $p_0(c)$ is the prior probability for each possible reason given based on their frequency or relative severity weights. Relational Graph Models (RGM), specifically relational Augmented Graphs (RAG), can improve retrieval accuracy by adding relational structures to retrieval methods, enabling them to identify connections among entities involved in an incident. This is why we have included the "path" term [6].

To constrain LLM behavior, the prompt includes explicit rules:

- The LLM should generate hypotheses supported by at least some portion of the retrieved evidence; if no supporting evidence is available, it should indicate uncertainty and suggest additional testing rather than providing conclusions.
- Each generated hypothesis must reference specific evidence elements (i.e., alarm code numbers, edges in the graph, etc.) that relate to that hypothesis.
- If there is conflicting evidence, the LLM should produce multiple competing hypotheses and identify specific actions that would help discriminate among them.

These constraints were chosen because hallucination research demonstrates that while LLMs can generate plausible-sounding (but erroneous) content, they cannot reliably detect hallucinated content. Detection and mitigation techniques will be necessary before they can be widely adopted in high-risk application areas [11].

2) Escalation Output Format: The output format is designed for immediate operational consumption. The output format contains:

- Probable Root Causes: Ranked list of probable root cause(s) along with qualitative or normalized-confidence bands associated with each cause.
- Evidence Trail: References to specific evidence elements, including alarm-code values, symptom descriptions, relevant graph-relations, and/or references to historical case (s).
- Recommended Next Actions: Prioritized recommendations for diagnostic/maintenance tasks and mitigation strategies to resolve the issue quickly.
- Draft Escalation Packet: Structured document summarizing key details (environment context, reproduction attempts, affected asset(s), timeline, attachments list).

This focus on generating output formatted to reflect a structurally-evidenced record of events is also representative of RAG's objective to create an evidentiary trail for inspectability and to enhance factuality by linking generation to external retrieval mechanisms [4].

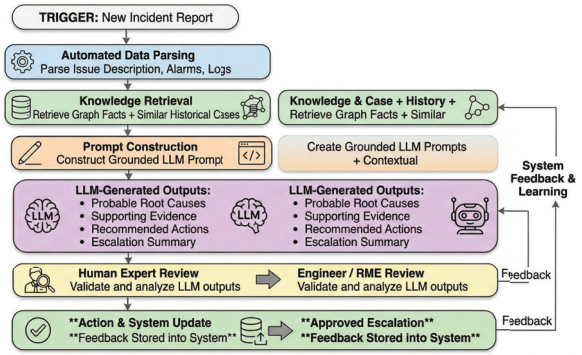


Fig. 2. Escalation Reasoning Workflow.

The human review stage is essential because: (i) the robotic FDD literature shows that the ambiguity of diagnostics arises from the uncertainty of its sensors and dynamic context [11]; (ii) there are many potential issues associated with LLMs being used as a diagnostic tool [2]; and (iii) all maintenance knowledge bases have to be maintained and updated incrementally and therefore they need to be curated constantly [6].

IV. Evaluation

This section outlines an evaluation methodology to compare the performance of the escalation engine against current workflows in terms of their diagnostic and operational effectiveness. As robotic maintenance data can often be proprietary, the evaluation will be defined as a retrospective off-line protocol and a pilot deployment protocol.

A. Experimental Setup

We recommend using a collection of historical incident data collected through the maintenance activities of real robots located at different sites and across multiple models. Each incident record should contain: the original ticket text and notes; alarm logs; any attachments or other relevant excerpts; the actions that were taken; and a confirmed root-cause label provided by the engineering team. This mirrors the literature on maintenance knowledge graphs, which states that a vast amount of historical maintenance data is available but remains largely unexploited due to its unstructured nature [9], [12].

To remove potential sources of confounders, incidents should be categorized based on:

- Frequent faults, with high numbers of precedents and consistent mapping.
- Novel faults, which have limited history, and represent one type of failure that has been identified as difficult to diagnose across multiple types of robots.
- Multi-hop faults. These fault types require reasoning across multiple system components and subsystems. For

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example, if a mechanical fault occurred and caused sensor readings to be out of range. Relational retrieval using GraphRAG has demonstrated its effectiveness in scenarios requiring multi-hop reasoning [1].

Additionally, there should be at least two classes of human evaluators. First, reliability and robotics engineers would be responsible for verifying the correctness of the top-ranked hypothesis explaining why an incident occurred and whether adequate supporting evidence was provided. Second would be support and maintenance engineers evaluating the clarity and usability of the escalated information. Socio-technical systems, including defined roles for all stakeholders and governing processes for implementing them, are also considered critical elements in managing the risk associated with the deployment of artificial intelligence systems [10].

B. Baseline Methods

A good assessment needs to have baselines that represent how your teams currently work:

- Manual Workflow: Developed by support engineers using their current applications and tools; there will be no application of automated retrieval past the ability to view basic dashboard views.
- Keyword Retrieval: Search of prior tickets and knowledge base articles by keyword(s); there will be no use of graph-based reasoning.
- Pure LLM: Generation of response from an LLM to the incident description or raw log entry without any specific indication of relationship to the knowledge graph.
- The Proposed Method: Use of the knowledge graph; retrieval of other cases that relate to the incident; constrained generation by an LLM.

Including a Pure LLM as one of the baselines is important because of documented issues of hallucination and unfaithfulness with unconstrained LLM usage, and because retrieval grounding is a major way to mitigate these risks [13].

C. Evaluation Metrics

These need to assess both the diagnostic quality of proposed solutions and their operational usability.

The diagnostic performance metrics include:

- Top-K Root Cause Accuracy: whether the engineer verifies that the root cause appears among the top K hypotheses outputted by the system;
- Mean Reciprocal Rank (MRR): a standard quality ranking metric for retrieval and recommendation-type problems, i.e., where systems produce ordered lists of hypotheses;
- Evidence Precision: fraction of cited evidence items that are valid and support each statement made.

The operational performance metrics include:

- Escalation Drafting Time: time required to create an escalation draft acceptable to the engineering team;
- Mean Time To Diagnosis (MTTD): time needed to develop an actionable set of hypotheses;
- Mean Time To Resolve (MTTR): overall time elapsed from opening an incident ticket until it was resolved, based on data collected during pilot deployments;
- Expert Acceptance Rate: proportion of system-generated escalations that were accepted after only minor editing.

As such, all of the above support the development of usable outputs for operational use rather than simply developing academic predictive capability. All of the above also fit within the scope of AI RMF guidance's emphasis on trust and governance.

D. Limitations

There are several fundamental limitations inherent to this work that should be explicitly recognized:

- Dependence on data quality and coverage: Maintenance operation data often exists in unstructured, incomplete, or inaccurate forms. Structured KG construction and ongoing incremental development are difficult and expensive, requiring continual quality control [5], [14].
- Domain-specific ontology effort: A robotics knowledge graph needs to accurately capture the structure of a robot's physical system and how it fails. Inaccurate representation of either will result in lower levels of explanation and lower retrieval relevance.
- Large Language Model (LLM) sensitivity and hallucination risk: Large language models can be grounded through retrieval, but component failures can occur due to hallucinations resulting from inadequate retrieval or contradictory context. Hallucination mitigation studies focused on Retrieval-Augmented Generation (RAG) provide ample evidence of such occurrences [3].
- Socio-technical deployment: Based on previous observations made by the Artificial Intelligence Risk Management Framework (AI RMF), we recognize that technical solutions alone cannot address the issues of trustworthiness and governance. Therefore, it is essential to establish governance structures with defined risk assessment procedures and accountability for risk management within an organization's AI system [6].

Understanding this socio-technical limitation has led us to continue revising and increasing the number of drafts for human review. Additionally, we have treated the engine as a tool to provide information, augmenting, but not replacing, the engineer's judgment.

V. Future Work

Future development will expand the prototype by adding enhanced sensor capabilities, continuous improvement, and generalization to robots at new locations.

One direction is multimodal expansion. In addition to textual fault descriptions and logs, future versions of the system could incorporate other signals such as vibration spectra, thermal images, acoustic emissions, and visual inspection images. Because PHM and maintenance systems often struggle to integrate multivariate data, a practical first step is to include summarized modalities, such as vibration vectors or

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anomaly-score summaries, as structured graph features rather than relying on raw multimodal interpretation. Prior work using VLMs on industrial troubleshooting diagrams suggests that such multimodal extraction may be useful, but current limitations indicate that deployment should remain human-supervised rather than fully autonomous [6], [13].

Another direction is continuous learning and broader generalization. The engine should improve over time by incorporating confirmed root causes, corrective actions, and escalation outcomes, with feedback loops used to update prior probabilities $p_0(c)$, revise graph relationships, and refine ontologies as engineers correct evidence mappings. Because KG pipelines require ongoing provenance-aware updates and governance, these improvements should be managed under AI RMF principles throughout the system life cycle [9]. Future work should also evaluate whether the same KG structure and retrieval pipeline generalize across different facilities, robot families, and product generations, potentially using alignment or embedding methods to map part taxonomies across platforms and testing performance on unseen sites.

VI. Conclusion

In order to make the process of root cause analysis and escalation development for advanced robotic systems easier, there exist several barriers, such as disparate pieces of information regarding a failure, failures across many different components within a system (interacting), and many of these failures can be dependent upon the way a system was designed/operated. Additionally, while much has been documented concerning issues with sensor dependency and dynamic operating environments associated with fault detection in robotics, a significant amount of work has also been conducted by researchers in the area of maintenance related to the need to formalize unstructured historical documentation into a format that can be used for reuse or knowledge representation [5].

This article presented a knowledge-graph and LLM-based escalation framework that consumes various forms of field data, creates a robot-fault knowledge graph, identifies relevant facts/evidence, and previous cases of similar faults, and produces lists of potential root causes and draft escalations from previously identified facts. This escalation framework leverages concepts of retrieval-augmented generation and graph-based retrieval to minimize unsupported reasoning, improve the ability to follow the chain of reasoning, and still allow for human-in-the-loop review, consistent with guidelines for managing risk in trustworthy AI [11].

The greater implications are that escalations become a formalizable, measurable, and improvable knowledge workflow; structured evidence/outcomes are created/reused/governed, thereby enabling scalable, intelligent maintenance and reliability engineering practices as the number of robotic deployments grows [10].

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